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Inventor(s) Marchek; Kyle et al.

CABLE REEL ASSEMBLIES AND SUB-ASSEMBLIES

Abstract

Cable reel assemblies and sub-assemblies are provided. A cable reel sub assembly includes a cable reel that has a first flange, a second flange, and a tubular body that extends between the first flange and the second flange. The first flange and the second flange each define a main opening and flange notches. The cable reel sub-assembly further includes a first pad and a second pad. Each of the first pad and the second pad define a central opening and pad notches. The cable reel sub-assembly further includes a first plug that extends through the central opening of the first pad and the main opening of the first flange. The cable reel sub-assembly further includes a second plug that extends through the central opening of the second pad and the main opening of the second flange.

Inventors: **Marchek; Kyle (Greer, SC), Reyes Hernandez; Raul (Nuevo Leon, MX)**

Applicant: **AFL Telecommunications LLC** (Duncan, SC)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a non-provisional application claiming the benefit of priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/562,447, filed Mar. 7, 2024, which is hereby incorporated by reference in its entirety.

FIELD

[0002] The present disclosure relates generally to improved reel assemblies for cables, such as optical fiber cables or other cables.

BACKGROUND

[0003] Reels are generally utilized for the transport and deployment of various products such as cables. In particular, electrical and optical fiber cables are frequently transported and deployed via reels. Typically, such cables are manufactured and stored on reels for transport to a customer, and the customer disposes of or returns the reel after deployment of the cable.

[0004] During the installation of cables in data centers, known cable reels must be removed from a shipping box. The reel requires reel holders for de-spooling the assemblies and additional locking equipment for selectively locking the reel. The installer spends time setting up the de-spooling equipment before the cable can be de-spooled from the reel. Additionally, if the reel is to be locked or restricted from movement, additional equipment is also necessary.

[0005] Accordingly, improved cable reel assemblies and sub-assemblies which meet the above requirements and address the above issues are desired and would be advantageous.

BRIEF DESCRIPTION

[0006] Aspects and advantages of the cable reel assemblies and sub-assemblies in accordance with the present disclosure will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the technology.

[0007] In accordance with one embodiment, a cable reel sub-assembly is provided. The cable reel sub assembly includes a cable reel that has a first flange, a second flange, and a tubular body that extends between the first flange and the second flange. The first flange and the second flange each define a main opening and flange notches. The cable reel sub-assembly further includes a first pad and a second pad. Each of the first pad and the second pad define a central opening and pad notches. The cable reel sub-assembly further includes a first plug that extends through the central opening of the first pad and the main opening of the first flange. The cable reel sub-assembly further includes a second plug that extends through the central opening of the second pad and the main opening of the second flange.

[0008] In accordance with another embodiment, a cable reel assembly is provided. The cable reel assembly includes a corrugated container that defines an interior and includes a flap. The cable reel assembly further includes a cable reel sub-assembly positioned within the interior of the corrugated container. The cable reel sub assembly includes a cable reel that has a first flange, a second flange, and a tubular body that extends between the first flange and the second flange. The first flange and the second flange each define a main opening and flange notches. The cable reel sub-assembly further includes a first pad and a second pad. Each of the first pad and the second pad define a central opening and pad notches. The cable reel sub-assembly further includes a first plug that extends through the central opening of the first pad and the main opening of the first flange. The cable reel sub-assembly further includes a second plug that extends through the central opening of the second pad and the main opening of the second flange.

[0009] In accordance with another embodiment, a method of locking a cable reel assembly is provided. The method includes moving a cable reel relative to a first pad and a second pad until a pad notch defined on one of the first pad and the second pad aligns with a flange notch defined on a

flange of the cable reel. The method further includes inserting a flap of a corrugated container into the pad notch and the flange notch to restrict movement between the cable reel and the first and second pads.
[0010] These and other features, aspects and advantages of the present cable reel assemblies and sub-assemblies will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the technology and, together with the description, serve to explain the principles of the technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0011] A full and enabling disclosure of the present cable reel assemblies and sub-assemblies, including the best mode of making and using the present systems and methods, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:
[0012] FIG. **1** illustrates a perspective view of a cable reel assembly in accordance with embodiments of the present disclosure;
[0013] FIG. **2** illustrates a perspective view of a cable reel sub-assembly in accordance with embodiments of the present disclosure;
[0014] FIG. **3** illustrates a perspective view of a cable reel in accordance with exemplary aspects of the present disclosure;
[0015] FIG. **4** illustrates a side view of a cable reel in accordance with exemplary aspects of the present disclosure;
[0016] FIG. **5** illustrates a side view of a pad coupled to a plug in accordance with exemplary aspects of the present disclosure;
[0017] FIG. **6** illustrates a perspective view of a plug in accordance with exemplary aspects of the present disclosure;
[0018] FIG. **7** illustrates a perspective view of a plug in accordance with exemplary aspects of the present disclosure;
[0019] FIG. **8** illustrates a perspective view of a cable reel assembly in accordance with embodiments of the present disclosure;
[0020] FIG. **9** illustrates a perspective view of a cable reel assembly in accordance with embodiments of the present disclosure;
[0021] FIG. **10** illustrates a perspective view of a cable reel in accordance with exemplary aspects of the present disclosure;
[0022] FIG. **11** illustrates an enlarged view of a tubular body with a key hole slot of a cable reel in accordance with exemplary aspects of the present disclosure; and
[0023] FIG. **12** is a flow chart of a method of locking a cable reel assembly in accordance with embodiments of the present disclosure.
DETAILED DESCRIPTION
[0024] Reference now will be made in detail to embodiments of the present cable reel assemblies and sub-assemblies, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation, rather than limitation of, the technology. In fact, it will be apparent to those skilled in the art that modifications and variations can be made in the present technology without departing from the scope or spirit of the claimed technology. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present disclosure covers such modifications and variations as come within the scope of the appended claims and their

equivalents.
[0025] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.
[0026] The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the invention. As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.
[0027] The term “fluid” may be a gas or a liquid. The term “fluid communication” means that a fluid is capable of making the connection between the areas specified.
[0028] The term “radially” refers to the relative direction that is substantially perpendicular to an axial centerline of a particular component, the term “axially” refers to the relative direction that is substantially parallel and/or coaxially aligned to an axial centerline of a particular component and the term “circumferentially” refers to the relative direction that extends around the axial centerline of a particular component.
[0029] Terms of approximation, such as “about,” “approximately,” “generally,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 1, 2, 4, 5, 10, 15, or 20 percent margin in either individual values, range(s) of values and/or endpoints defining range(s) of values. When used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction. For example, “generally vertical” includes directions within ten degrees of vertical in any direction, e.g., clockwise or counter-clockwise.
[0030] The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein. As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of features is not necessarily limited only to those features but may include other features not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive- or and not to an exclusive-or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).
[0031] Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.
[0032] Referring now to the drawings, FIGS. **1** through **9** illustrate a cable reel assembly **10** and components thereof in accordance with embodiments of the present disclosure. A cable reel assembly **10** in accordance with the present disclosure may advantageously be utilized to transport and deploy cables, such as electrical or fiber optic cables or other cables. Particularly, FIG. **1** illustrates a perspective view of a cable reel assembly **10**; FIG. **2** illustrates a perspective view of a

cable reel sub-assembly **12**; FIG. **3** illustrates a perspective view of a cable reel **14**; FIG. **4** illustrates a side view of a cable reel **14**; FIG. **5** illustrates a side view of a pad **54** connected to a plug **56**; FIG. **6** illustrates a first perspective view of the plug **56**; FIG. **7** illustrates a second perspective view of the plug **56**; and FIGS. **8** and **9** each illustrate a perspective view of a cable reel assembly **10** in a restricted position.

[0033] The cable reel assembly **10** (FIG. **1**) may include a corrugated container **20** that defines an interior **22** (FIGS. **8** and **9**) and includes one or more flaps **24**A, **24**B, **26**A, **26**B (FIGS. **8** and **9**). Particularly, the corrugated container **20** may include one or more side walls **28**, and the one or more flaps **24**A, **24**B, **26**A, **26**B may each extend from one or more side walls **28** to a terminal end. The one or more side walls **28** may be mutually perpendicular to one another. As shown in FIGS. **8** and **9**, the corrugated container **20** may include inner flaps **24**A, **24**B (such as a first inner flap **24**A and a second inner flap **24**B) and outer flaps **26**A, **26**B (such as a first outer flap **26**A and a second outer flap **26**B). The inner flaps **24**A, **24**B may have a smaller width than the outer flaps **26**A, **26**B. The outer flaps **26**A, **26**B and the inner flaps **24**A, **24**B may be pivotable relative to the side walls **28** from which they extend to provide for selective access to the interior **22**. For example, the outer flaps **26**A, **26**B and the inner flaps **24**A, **24**B may be movable to an open position that provides an opening to access the interior **22** of the corrugated container **20** (as shown in FIGS. **8** and **9**).

[0034] In various embodiments, the corrugated container **20** may be one of a paper corrugated container (e.g., cardboard) or a plastic corrugated container. In this way, the corrugated container **20** may be formed from a compliant material, such that the flaps **24**A, **24**B, **26**A, **26**B may be bendable and/or flexible. In various embodiments, the corrugated container **20** may be one of a regular slotted container (RSC), a half slotted container (HSC), an overlap slotted container (OSC), a full overlap slotted container (FOL), A center special slotted container (CSSC), or other suitable corrugated container. The corrugated container **20** may be formed from a single sheet of corrugated material (such as paper, cardboard, or plastic) that is cut and scored to form the corrugated container **20** having the flaps **24**A, **24**B, **26**A, **26**B that provide for selective access to the interior **22**.

[0035] In many embodiments, the cable reel **14**, including the flanges **38** and the tubular body **42**, may be formed from a corrugated material (such as a corrugated paper or corrugated plastic). In various embodiments, the cable reel **14** may be formed from corrugated material (such as corrugated paper or corrugated plastic). Forming both the cable reel **14** and the corrugated container from paper and/or plastic advantageously allows for the components to be recycled once the deployment of a cable is complete.

[0036] As shown in FIG. **8**, the inner flaps **24**A, **24**B may extend the full width of the corrugated container but may not extend the full length of the corrugated container **20**. For example, the inner flaps **24**A, **24**B may each define a length L**1** and a width W**1** mutually perpendicular to one another. The length L**1** may be defined from a base **30** (e.g., where the inner flap **24**A, **24**B intersects with the side wall **28** of the corrugated container **20**) of the inner flap **24**A, **24**B to a terminal end **32** of the inner flap **24**A, **24**B. The length L**1** may be less than a length of the corrugated container **20**, such that the inner flaps **24**A, **24**B do not overlap and do not collectively enclose the interior **22** when in a folded position. The width W**1** may extend the entire width of the corrugated container **20** (e.g., between two side walls **28**).

[0037] The outer flaps **26**A, **26**B may extend the full length and the full width of the corrugated box **20**. For example, the outer flaps **26**A, **26**B may each define a length L**2** and a width W**2** mutually perpendicular to one another. The length L**2** may be defined from a base **34** (e.g., where the outer flap **26**A, **26**B intersects with the side wall **28** of the corrugated container **20**) of the outer flap **26**A, **26**B to a terminal end **36** of the outer flap **26**A, **26**B. The length L**2** may be generally equal to a width of the corrugated container **20**, such that the outer flaps **26**A, **26**B extend across the entire width of the corrugated container **20**, overlap with one another, and enclose the interior **22** when in a folded position. The width W**2** may extend the entire length of the corrugated container **20** (e.g., between two side walls **28**).

[0038] Referring back to FIGS. **2** and **3**, a cable reel sub-assembly **12**, which includes a cable reel **14**, may be included in the cable reel assembly **10**. For example, as shown in FIGS. **8** and **9**, the cable reel-sub assembly **12** may be positioned within the interior **22** of the corrugated container **20**. The cable reel sub assembly **12** may be selectively removable from the interior **22** by adjusting the flaps **24**A, **24**B, **26**A, **26**B to an open position to create an opening to the interior **22** and sliding the cable reel sub-assembly **12** out of the opening.

[0039] As shown in FIGS. **2** and **3**, the cable reel sub-assembly **12** may include the cable reel **14**. The cable reel **14** may include flanges **38** and a tubular body **42** extending between the flanges **38**. Particularly, the cable reel **14** may include a first flange **38**, a second flange **38**, and a tubular body **42** extending between the first flange **38** and the second flange **38**. The flanges **38** may each define a periphery **44** (or edge) that is round (e.g., a circle, oval, or other shape having a continuous curvature). The tubular body **42** may extend between the first flange **38** and the second flange **38**, and the tubular body **42** may have a diameter that is less than a diameter of the first flange **38** and the second flange **38**. In operation, a cable may be wound about the tubular body **42** between the flanges **38**. In this way, the flanges **38** having a larger diameter than the tubular body **42** is advantageous because it prevents the cable from sliding off the tubular body **42**.

[0040] As shown in FIG. **3**, the flanges **38** may each define a main opening **48** (only one shown). The main opening **48** may be circular (however, other shapes may be possible). The main opening **48** may be centered on each of the first flange **38** and the second flange **38** (e.g., geometrically centered). For example, the first flange **38** and the second flange **38** may be circularly shaped, and the main opening **48** may be centered on the circle, which advantageously provides for uniform rotation of the cable reel **14** when winding or unwinding cable from the tubular body **42**.

[0041] Additionally, each of the first flange **38** and the second flange **38** may define flange notches **50**, **52**. Particularly, the first flange **38** and the second flange **38** may each define a pair of outer flange notches **50** and a pair of inner flange notches **52**. The pair of outer flange notches **50** and the pair of inner flange notches **52** may each extend from a respective periphery **44** of a respective flange **38**. For example, for the first flange **38**, the pair of outer flange notches **50** may extend from the first periphery **44**, and the pair of inner flange notches **52** may extend from the first periphery **44**. Similarly, for the second flange **38**, the pair of outer flange notches **50** may extend from the second periphery **44**, and the pair of inner flange notches **52** may extend from the second periphery **44**.

[0042] The pair of outer flange notches **50** may be spaced apart from one another and disposed on opposite sides of the main opening **48**. In other words, the main opening **48** may be disposed between (e.g., directly between) the pair of outer flange notches **50**. That is, the main opening **48** may be disposed equidistant from each outer flange notch in the pair of outer flange notches **50**. Similarly, the pair of inner flange notches **52** may be spaced apart from one another and disposed on opposite sides of the main opening **48**. In other words, the main opening **48** may be disposed between (e.g., directly between) the pair of inner flange notches **52**. That is, the main opening **48** may be disposed equidistant from each inner flange notch in the pair of outer flange notches **52**.

[0043] Referring now to FIG. **2**, the cable reel sub-assembly **12** may further include pads **54** and plugs **56** that rotatably couple the pads **54** to the respective flange **38**. For example, the cable reel sub-assembly **12** may include a first pad **54** and a second pad **54** spaced apart from one another and each disposed adjacent to a respective flange **38**. For example, the first pad **54** may be coupled to the first flange **38** via a first plug **56**, and the second pad **54** may be coupled to the second flange **38** via a second plug **56**. The pads **54** may each define a central opening **58**. When assembled, as shown in FIG. **2**, a plug **56** may be extend through the central opening **58** of a pad **54** and the main opening **48** of a flange **38** to couple the pad **54** to the flange **38**. More specifically, a first plug **56** may extend through the first central opening **58** of the first pad **54** and the first main opening **48** of the first flange **38** to couple the first pad **54** to the first flange **38**. Similarly, the second plug **56** may extend through the second central opening **58** of the second pad **54** and the second main opening **48** of the second flange **38** to couple the second pad **54** to the second flange **38**. The cable reel **14** may be rotatable about the plugs **56** relative to the pads **54** to spool or unspool cable as desired.

[0044] The pads **54** may each define pad notches **60**, **62** that are alignable with the flange notches **50**, **52** (as shown in FIG. **2**) by rotating the cable reel **14** about the plugs **56** relative to the pads **54**. Specifically, the pads **54** may each define a pair of outer pad notches **60** and a pair of inner pad notches **62**. The pair of outer pad notches **60** may be alignable with the pair of outer flange notches **50**. More specifically, the pair of outer pad notches **60** in the first pad **54** may be alignable with the pair of outer flange notches **50** in the first flange **38**, and the pair of outer pad notches **60** in the second pad **54** may be alignable with the pair of outer flange notches **50** in the second flange **38**. The pair of inner pad notches **62** may be alignable with the pair of inner flange notches **52**. More specifically, the pair of inner pad notches **62** in the first pad **54** may be alignable with the pair of inner flange notches **52** in the first flange **38**, and the pair of inner pad notches **62** in the second pad **54** may be alignable with the pair of inner flange notches **52** in the second flange **38**.

[0045] The first pad **54** and the second pad **54** may each define an edge **64** (at which the pads terminate). In the exemplary embodiment shown, the pads **54** may be shaped as a square or rectangle (e.g., having four edges, which may be equally sized). However, in other embodiments, the pads **54** may have another shape, such as any polygonal shape (having any number of edges). In the embodiment shown, the first pad **54** and the second pad **54** may each include an outer edge **66**, an inner edge **68**, and side edges **70** extending between the outer edge **66** and the inner edge **68**. The pad notches may. The pad notches **60**, **62** may extend from the edge **64** of each of the first pad **54** and the second pad **54**. Specifically, the outer pad notches **60** may extend from the outer edge **66**, and the inner pad notches **62** may extend from the inner edge **68**.

[0046] As shown in FIG. **3**, the flanges **38** may each define a plurality of slots **71** extending about the main opening **48**. In such embodiments, the tubular reel **42** may include a plurality of foldable tabs **72** that each extend through a respective slot of the plurality of slots **71** to couple the flanges **38** to the tubular reel **42**. However, other means of coupling the tubular reel **42** to the flanges **38** may be possible and the present disclosure should not be limited to one particular means for coupling these components unless specifically recited in the claims.

[0047] Referring now specifically to FIGS. **4** and **5**, a side view of a cable reel **14** (FIG. **4**) and a side view of a pad **54** coupled to a plug **56** (FIG. **5**) are illustrated in accordance with exemplary aspects of the present disclosure. As shown, the cable reel **14** may include a flange **38** having a circular periphery **44**. The main opening **48** may be defined in a center of the flange **38**. The flange **38** may define a plurality of slots **71** that at least partially surround the main opening **48**. A plurality of foldable tabs **72** of the tubular reel **42** may extend through the plurality of slots **71** to couple the tubular reel to the flange **38**.

[0048] As shown in FIG. **4**, in some embodiments, the flange notches **50** may each extend from the periphery **44** towards the main opening **48** to an end **74**. That is, the flange notches **50** may be angled (e.g., relative to a vertical direction V, which may be aligned with the direction of gravity) such that they extend towards from the periphery **44** towards the main opening **48** to the end **74**. In other embodiments, as shown in FIGS. **2** and **3**, the flange notches **50** may extend entirely vertically (e.g., parallel to the side edges **70** and perpendicular to the inner edge **68** and the outer edge **66**).

[0049] As shown in FIG. **5**, the pad **54** may be shaped generally rectangularly shaped (such as a square in many embodiments) having edges **64**. Particularly, the pad **54** may include an outer edge **66** and an inner edge **68** spaced apart from one another. Side edges **70** may extend between the outer edge **66** and the inner edge **68**. A central opening **58** may be defined through a center of the pad **54**, and the plug **56** may extend through the central opening **58** (and into the main opening **48** of the flange **38** when fully assembled).

[0050] As shown in FIG. **5**, the pad notches **60** may each extend from the edge **64** (such as the outer edge **66**) towards the central opening **58** to an end **76**. That is, the pad notches **60** may be angled (e.g., relative to a vertical direction V, which may be aligned with the direction of gravity) such that they extend towards from the edge **64** towards the central opening **58** to the end **76**. In other embodiments, as shown in FIGS. **2** and **3**, the flange notches **50** may extend entirely vertically (e.g., parallel to the side edges **70** and perpendicular to the inner edge **68** and the outer edge **66**).

[0051] Additionally, as shown in FIGS. **4** and **5**, in many embodiments, the flange **38** and the pad **54** may each define a fastener hole through which a fastener may extend to lock the cable reel **14** from movement relative to the pad **54**. For example, at least one of the first pad **54** and the second pad **54** may define a pad fastener hole **78** (FIG. **5**), and at least one of the first flange **38** and the second flange **38** may define a flange fastener hole **80** (FIG. **4**) that is alignable with the pad fastener hole **78**. The pad fastener hole **80** may be disposed between the flange notches **50** closer to the periphery **44** than the main opening **48**. Similarly, the pad fastener hole **78** may be disposed between the pad notches **60** closer to an edge **64** (e.g., the outer edge **66**) than the central opening **58**.

[0052] When the pad **54** is coupled to the flange **38** via the plug **56**, the cable reel **14** may be rotated about the plug **56** relative to the pad **54** until the pad fastener hole **78** aligns with the flange fastener hole **80**. Subsequently, a fastener may be inserted through the pad fastener hole **78** and the flange fastener hole **80** to restrict (or entirely prevent) the cable reel **14** from moving relative to the pad **54**. In exemplary implementations, the cable wound about the cable reel **14** may be wrapped in a poly tube plastic. Excess poly tube plastic may be forced through the holes **78**, **80** to lock the cable reel **14**. In other implementations, a fastener (such as a bolt, screw, zip tie, or other fastener) may be inserted through the holes **78**, **80** to lock the cable reel **14** relative to the pad **54**.

[0053] Referring now specifically to FIGS. **6** and **7**, two different perspective views of a plug **56** are illustrated in accordance with exemplary aspects of the present disclosure. As shown, the plug **56** may include an outer ring **84** and an inner ring **86** spaced apart from the outer ring **84**. A plurality of ribs **88** may extend between the outer ring **84** and the inner ring **86**. The outer ring **84** and the inner ring **86** may extend between a base **90** and a tip **92**. A lip **94** may extend outwardly (e.g., radially outwardly) from the outer ring **84** at the tip **92**.

[0054] Referring now specifically to FIGS. **8** and **9**, two perspective views of a cable reel assembly **10** are illustrated in accordance with embodiments of the present disclosure. As shown, the cable reel sub-assembly **12** may be movable between a restricted position (FIGS. **8** and **9**) and an unrestricted position (FIG. **2**). Particularly, as shown in FIGS. **8** and **9**, a flap **24**A, **24**B may extend into the flange notches **50** and the pad notches **60** such that the cable reel **14** is restricted from movement relative to the first pad **54**, the second pad **54**, and the corrugated container **20** when in the restricted position. For example, the inner flap(s) **24**A, **24**B may extend into the flange notches **50** (in both the first and second flanges **38**) and the pad notches **60** (in first and second pads **54**), which restricts rotational movement of the flanges **38** relative to the pads **54**, and which restricts translational movement of the cable reel sub-assembly **12** relative to the corrugated container **20**. By contrast, when in the unrestricted position (as shown in FIG. **2**), the cable reel may be permitted to move (e.g., rotate about the plugs **56**) relative to the first pad **54** and the second pad **54**.

[0055] Referring back to FIG. **2** briefly, the cable reel **14** is rotatable about the first plug **56** and the second plug **56** relative to the first pad **54** and the second pad **54** between an aligned position (shown in FIG. **2**) and an unaligned position. The pad notches **60** are aligned with the flange notches **50** in the aligned position as shown in FIGS. **2**, **8**, and **9**. When in the unaligned position, the cable reel **14** may be rotated relative to the pads **54** such that the flange notches **50** are not aligned with the pad notches **60**. Referring now to FIGS. **8** and **9**, the inner flaps **24**A, **24**B may be removably insertable into the flange notches **50** and the pad notches **60** when the cable reel **14** is in the aligned position. This advantageously locks the cable reel **14** relative to the pads **54** and prevents the cable reel sub-assembly **12** from being withdrawn from the interior **22**.

[0056] Referring now to FIGS. **10** and **11**, a cable reel **14** and a tubular body **42** are illustrated. Specifically, FIG. **10** illustrates an enlarged perspective view of a cable reel **14**, and FIG. **11** illustrates an enlarged view of a portion of a tubular body **42** of the cable reel **14**, in accordance with various exemplary aspects of the present disclosure. The cable reel **14** may include flanges **38** and a tubular body **42** extending between, and coupled to, the flanges **38**. The tubular body **42** may define a keyhole opening **96**. The keyhole opening **96** may be disposed closer to one of the flanges **38** than the other of the flanges **38** (e.g., closer to the first flange than the second flange). As shown in FIG. **11**, the keyhole **96** may include an opening portion **98** and a slot portion **100** extending from the opening portion **98**. The slot portion **100** may have a longer length than the opening portion **98** and a shorter width than the opening portion **98**. In operation, a cable (and/or a portion of poly plastic tube that surrounds the cable) may be inserted into the opening portion **98** of the keyhole slot **96** and slid down the slot portion **100** of the keyhole slot **96**. This advantageously prevents the cable from being entirely unspooled from the cable reel **14**.
[0057] Referring now to FIG. **12**, a flow diagram of one embodiment of a method **1200** of locking a cable reel assembly is illustrated in accordance with embodiments of the present subject matter. In general, the method **1200** will be described herein with reference to the cable reel assembly **10**, the cable reel sub-assembly **12**, and the cable reel **14** described above with reference to FIGS. **1-11**. However, it will be appreciated by those of ordinary skill in the art that the disclosed method **1200** may generally be utilized with any other suitable system configuration. In addition, although FIG. **12** depicts steps performed in a particular order for purposes of illustration and discussion, the methods discussed herein are not limited to any particular order or arrangement unless otherwise specified in the claims. One skilled in the art, using the disclosures provided herein, will appreciate that various steps of the methods disclosed herein can be omitted, rearranged, combined, and/or adapted in various ways without deviating from the scope of the present disclosure. Dashed boxes indicate optional steps of the method **1200**.
[0058] In exemplary implementations, the method **1200** may include at (**1202**) moving a cable reel relative to a first pad and a second pad until a pad notch defined on one of the first pad and the second pad aligns with a flange notch defined on a flange of the cable reel. In certain implementations, moving at (**1202**) may further include at (**1204**) rotating the cable reel about a first plug and a second plug relative to the first pad and the second pad until the pad notch aligns with the flange notch. That is, the cable reel sub-assembly may be movable (or rotatable) between an unaligned position and an aligned position in which the pad notch overlaps with the flange notch to form a single continuous notch into which the flap may be inserted. In many implementations, the method **1200** may further include at (**1206**) inserting a flap of a corrugated container into the pad notch and the flange notch to restrict (or entirely prevent) movement between the cable reel and the first and second pads.
[0059] The cable reel assembly **10** described hereinabove advantageously facilitates the installation of a cable without additional equipment. That is, no de-spooling equipment or cable reel locking equipment is needed. Once the cable is installed, the cable reel assembly **10** can be quickly knocked down and shipped back to the supplier or placed into a recycling container.
[0060] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.
[0061] Further aspects of the invention are provided by the subject matter of the following clauses:
[0062] A cable reel sub-assembly comprising: a cable reel having a first flange, a second flange, and a tubular body extending between the first flange and the second flange, wherein the first

flange and the second flange each define a main opening and flange notches; a first pad and a second pad, each of the first pad and the second pad defining a central opening and pad notches; a first plug extending through the central opening of the first pad and the main opening of the first flange; and a second plug extending through the central opening of the second pad and the main opening of the second flange.
[0063] The cable reel sub-assembly as in any preceding clause, wherein the cable reel is rotatable about the first plug and the second plug relative to the first pad and the second pad between an aligned position and an unaligned position, wherein the pad notches are aligned with the flange notches in the aligned position.
[0064] The cable reel sub-assembly as any preceding clause, wherein the first pad and the second pad each define an edge, wherein the pad notches defined in the first pad and the second pad each extend from a respective edge.
[0065] The cable reel sub-assembly as in any preceding clause, wherein the first flange and the second flange each define a periphery, and wherein the flange notches defined in the first flange and the second flange each extend from a respective periphery.
[0066] The cable reel sub-assembly as in any preceding clause, wherein at least one of the first pad and the second pad define a pad fastener hole, and wherein at least one of the first flange and the second flange define a flange fastener hole that is alignable with the pad fastener hole.
[0067] The cable reel sub-assembly as in any preceding clause, wherein the tubular body defines a keyhole having an opening portion and a slot portion extending from the opening portion.
[0068] A cable reel assembly comprising: a corrugated container defining an interior and including a flap; a cable reel sub-assembly positioned within the interior of the corrugated container, the cable reel sub-assembly comprising: a cable reel having a first flange, a second flange, and a tubular body extending between the first flange and the second flange, wherein the first flange and the second flange each define a main opening and flange notches; a first pad and a second pad, each of the first pad and the second pad defining a central opening and pad notches; a first plug extending through the central opening of the first pad and the main opening of the first flange; and a second plug extending through the central opening of the second pad and the main opening of the second flange.
[0069] The cable reel assembly as in any preceding clause, wherein the corrugated container is one of a paper corrugated container and a plastic corrugated container.
[0070] The cable reel assembly as in any preceding clause, wherein the cable reel sub-assembly is movable between a restricted position and an unrestricted position within the corrugated container.
[0071] The cable reel assembly as in any preceding clause, wherein the flap extends into the flange notches and the pad notches such that the cable reel is restricted from movement relative to the first pad and the second pad when in the restricted position.
[0072] The cable reel assembly as in any preceding clause, wherein the cable reel is permitted to move relative to the first pad and the second pad when in the unrestricted position.
[0073] The cable reel assembly as in any preceding clause, wherein the cable reel is rotatable about the first plug and the second plug relative to the first pad and the second pad between an aligned position and an unaligned position, wherein the pad notches are aligned with the flange notches in the aligned position.
[0074] The cable reel assembly as in any preceding clause, wherein the flap is removably insertable into the flange notches and the pad notches when the cable reel is in the aligned position.
[0075] The cable reel assembly as in any preceding clause, wherein the first pad and the second pad each define an edge, wherein the pad notches defined in the first pad and the second pad each extend from a respective edge.
[0076] The cable reel assembly as in any preceding clause, wherein the first flange and the second flange each define a periphery, and wherein the flange notches defined in the first flange and the second flange each extend from a respective periphery.

[0077] The cable reel assembly as in any preceding clause, wherein at least one of the first pad and the second pad define a pad fastener hole, and wherein at least one of the first flange and the second flange define a flange fastener hole that is alignable with the pad fastener hole.
[0078] The cable reel assembly as in any preceding clause, wherein the tubular body defines a keyhole having an opening portion and a slot portion extending from the opening portion.
[0079] A method of locking a cable reel assembly, the method comprising: moving a cable reel relative to a first pad and a second pad until a pad notch defined on one of the first pad and the second pad aligns with a flange notch defined on a flange of the cable reel; and inserting a flap of a corrugated container into the pad notch and the flange notch to restrict movement between the cable reel and the first and second pads.
[0080] The method as in any preceding clause, wherein moving the cable reel further comprises: rotating the cable reel about a first plug and a second plug relative to the first pad and the second pad until the pad notch aligns with the flange notch.

Claims

1. A cable reel sub-assembly comprising: a cable reel having a first flange, a second flange, and a tubular body extending between the first flange and the second flange, wherein the first flange and the second flange each define a main opening and flange notches; a first pad and a second pad, each of the first pad and the second pad defining a central opening and pad notches; a first plug extending through the central opening of the first pad and the main opening of the first flange; and a second plug extending through the central opening of the second pad and the main opening of the second flange.
2. The cable reel sub-assembly as in claim 1, wherein the cable reel is rotatable about the first plug and the second plug relative to the first pad and the second pad between an aligned position and an unaligned position, wherein the pad notches are aligned with the flange notches in the aligned position.
3. The cable reel sub-assembly as in claim 1, wherein the first pad and the second pad each define an edge, wherein the pad notches defined in the first pad and the second pad each extend from a respective edge.
4. The cable reel sub-assembly as in claim 1, wherein the first flange and the second flange each define a periphery, and wherein the flange notches defined in the first flange and the second flange each extend from a respective periphery.
5. The cable reel sub-assembly as in claim 1, wherein at least one of the first pad and the second pad define a pad fastener hole, and wherein at least one of the first flange and the second flange define a flange fastener hole that is alignable with the pad fastener hole.
6. The cable reel sub-assembly as in claim 1, wherein the tubular body defines a keyhole having an opening portion and a slot portion extending from the opening portion.
7. A cable reel assembly comprising: a corrugated container defining an interior and including a flap; a cable reel sub-assembly positioned within the interior of the corrugated container, the cable reel sub-assembly comprising: a cable reel having a first flange, a second flange, and a tubular body extending between the first flange and the second flange, wherein the first flange and the second flange each define a main opening and flange notches; a first pad and a second pad, each of the first pad and the second pad defining a central opening and pad notches; a first plug extending through the central opening of the first pad and the main opening of the first flange; and a second plug extending through the central opening of the second pad and the main opening of the second flange.
8. The cable reel assembly as in claim 7, wherein the corrugated container is one of a paper corrugated container and a plastic corrugated container.
9. The cable reel assembly as in claim 7, wherein the cable reel sub-assembly is movable between a

restricted position and an unrestricted position within the corrugated container.

10. The cable reel assembly as in claim 9, wherein the flap extends into the flange notches and the pad notches such that the cable reel is restricted from movement relative to the first pad and the second pad when in the restricted position.

11. The cable reel assembly as in claim 9, wherein the cable reel is permitted to move relative to the first pad and the second pad when in the unrestricted position.

12. The cable reel assembly as in claim 7, wherein the cable reel is rotatable about the first plug and the second plug relative to the first pad and the second pad between an aligned position and an unaligned position, wherein the pad notches are aligned with the flange notches in the aligned position.

13. The cable reel assembly as in claim 12, wherein the flap is removably insertable into the flange notches and the pad notches when the cable reel is in the aligned position.

14. The cable reel assembly as in claim 7, wherein the first pad and the second pad each define an edge, wherein the pad notches defined in the first pad and the second pad each extend from a respective edge.

15. The cable reel assembly as in claim 7, wherein the first flange and the second flange each define a periphery, and wherein the flange notches defined in the first flange and the second flange each extend from a respective periphery.

16. The cable reel assembly as in claim 7, wherein at least one of the first pad and the second pad define a pad fastener hole, and wherein at least one of the first flange and the second flange define a flange fastener hole that is alignable with the pad fastener hole.

17. The cable reel assembly as in claim 8, wherein the tubular body defines a keyhole having an opening portion and a slot portion extending from the opening portion.

18. A method of locking a cable reel assembly, the method comprising: moving a cable reel relative to a first pad and a second pad until a pad notch defined on one of the first pad and the second pad aligns with a flange notch defined on a flange of the cable reel; and inserting a flap of a corrugated container into the pad notch and the flange notch to restrict movement between the cable reel and the first and second pads.

19. The method as in claim 18, wherein moving the cable reel further comprises: rotating the cable reel about a first plug and a second plug relative to the first pad and the second pad until the pad notch aligns with the flange notch.